1.59 Large Data Set: AQA

Worked Solutions

1 a)

BMW, Ford, Toyota, Vauxhall, Volkswagen

1 b)

2002, 2016

1 c)

London, North West, South West

1 d)

2002, 2016

1 e)

Petrol, Diesel, Electric, Gas/Petrol, Electric/Petrol

1 f)

2 door saloon, 4 door saloon, Saloon, Convertible, Coupe, Estate, 3 Door Hatchback, 5 Door Hatchback, Multi Purpose Vehicle

2 a)

g/km

2 b)

g/km

2 c)

g/km

2 d)

g/km

2 e)

g/km

2 f)

kg

2 g)

cubic cm

3)

There is data missing from the LDS for carbon monoxide, nox, particulate and hydrocarbon emissions.

4)

Mass: 0 kg is a mistake as every car has a mass as it should include the 75 kg for driver Particulate emissions: The LDS does have blanks in particulate emissions where it has not been recorded but there are no 0 Engine size: There is one car with a 0 for engine size in the LDS

5 a)

Systematic sampling

5 b)

Quota sampling

5 c)

Stratified sampling

6 a)

Incorrect as 75kg is included in the mass, not the mean mass of the population at the time

6 b)

Incorrect as the LDS doesn't include: All car manufacturers All regions of the UK Vehicles other than cars

7 a)

Mean is 1405.4 cm^3, sample standard deviation is 334.4 cm^3 (population standard deviation is 317.2 cm^3)

7 b)

There are significantly more cars in the sample from London than expected as London is the smallest number of cars by region in the LDS (998 from London, 1164 from the North West and 1665 from the South West)

7 c)

$1405.4 - 2 \times 334.4 = 736.6$ cm^3 $1405.4 + 2 \times 334.4 = 2074.2$ cm^3 (if using population: $1405.4 - 2 \times 317.2 = 771$ cm^3 $1405.4 + 2 \times 317.2 = 2039.8$ cm^3) The BMW with engine size 2171cm^3 is an outlier.

7 d)

The engine size is larger on average in the whole data set than the sample.

7 e)

The engine size is more spread out on average in the whole data set than the sample.

8 a)

2002: $\frac{46335}{302} = 153.427\ldots153$ g/km

2016: $\frac{127096}{1097} = 115.8577\ldots116$ g/km

8 b)

$$\bar{x}_{\text{all}} = 120.37 \text{ g/km} = \frac{127\,096 + \sum x_{D'}}{2529}$$

$$\sum x_{D'} = 177319.73$$

so $\bar{x}_{D'} = \frac{\sum x_{D'}}{2529-1097} = 123.82\ldots124$g/km

8 c)

There's only one electric vehicle in the large data set, and its CO_2 emissions are zero.

9 a)

$(2075 - 1693) \times 8 = 3056$ represents 8 cars

Factor is $8 \div 3056 = \frac{1}{382}$

$$(929 - 738) \times 2 \times \frac{1}{382} = 1$$

$$(1311 - 929) \times 16 \times \frac{1}{382} = 16$$

$$(1502 - 1311) \times 32 \times \frac{1}{382} = 16$$

$$(1693 - 1502) \times 18 \times \frac{1}{382} = 9$$

$$(2075 - 1693) \times 8 \times \frac{1}{382} = 8$$

Total of 50 cars were sampled.

9 b)

Mean:

$$\frac{382 \times 833.5 + 6112 \times 1120 + 6112 \times 1406.5 + 3438 \times 1597.5 + 3056 \times 1884}{382 + 6112 + 6112 + 3438 + 3056}$$

$$= \frac{27010074}{19100} = 1414.14 \text{ kg}$$

Or

$$\text{Mean} = \frac{1 \times 833.5 + 16 \times 1120 + 16 \times 1406 + 9 \times 1597.5 + 8 \times 1884}{50}$$

$$= \frac{70707}{50} = 1414.14 \text{ kg}$$

9 c)

$$(191 \times 16 + 191 \times 32 + 191 \times 18) \times \frac{1}{382} = 33 \text{ cars}$$

Therefore $P(1120 < \text{mass} < 1693) = \frac{33}{50} = 0.66$

10 a)

$k = \frac{26}{125}$

10 b)

Male: 100 Female: 52 Unknown: 8 Company: 90

10 c)

The proportions of Male/female/unknown and company are very similar to that of the large data set.

11)

Engine sizes vary between 647 and 4951 so the data lies within this, but the middle 50% of the Large Data Set lies between 1395 and 1968. The box plot has a much lower middle 50% so it is likely that there has been a higher selection of lower engines selected which indicates it may not have been sampled randomly.

12 a)

Mean: $\frac{5592}{40} = 139.8$ g/km

Standard deviation: $\sqrt{\frac{\sum x^2 - \frac{(\sum x)^2}{n}}{n-1}} = \sqrt{\frac{802846 - \frac{(5592)^2}{40}}{40-1}} = 23.25$ g/km

12 b)

$$X \sim \mathrm{N}(139.8, 23.25^2)$$

$$\mathrm{P}(X > 150) = 0.3304$$

12 c)

$\mathrm{P}(X = 114) = 0$